SA 220 AQUACULTURE

CLASSIFICATION OF AQUACULTURE

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- Aquaculture can be classified based on the following;

A. Feeding and levels of intensification in aquaculture

1.**Extensive aquaculture:** Natural food as only source; sometimes it may also involve fertilisation.

- Large ponds ranging 1 to 5 ha in area are used for farming and Harvest ranges between 500 to 2000 kgs /ha.

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2. **Semi-intensive aquaculture:** Natural food with fertilisation and, when yield increases, supplemental feeding (incomplete feeds) is used and size ranges between 0.5 to 1 hectare.

- **Feed of often less than optimum quality and/or quantity** is added for direct consumption to **supplement natural foods** - feeds are usually nutritionally incomplete. EG Using maize bran
- Harvest – 2500 to 10000 kgs (Fish)

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3. **Intensive aquaculture:** Use of nutritionally balanced diets ('complete diets') and minor or no inputs through natural food.



- Usually water is control for instance aeration will be done.
- Production and sizes are higher than the other two categories.



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B. Farming techniques

1. Monoculture (Single species)
2. Polyculture (Several species)
3. Integration of aquaculture and agriculture (IAA



CLASSIFICATION OF AQUACULTURE

C. CULTURE FACILITY

1. Pond culture
2. Cage culture
3. Pen culture
4. Tank culture







CLASSIFICATION OF SALINITY

D. Salinity

1. **Freshwater:** Freshwater contains low levels of dissolved salts, typically less than 0.5 parts per thousand (ppt) or 0.5‰ salinity.

- It includes water found in lakes, rivers, streams, and groundwater.

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2. Brackish Water: Brackish water has a moderate salinity level, higher than freshwater but lower than seawater.

- It typically contains between 0.5 to 30 ppt salinity.
- Commonly found in estuaries, where freshwater from rivers mixes with seawater from the ocean.
- It can also occur in coastal lagoons, salt marshes, and certain groundwater sources.

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3. Seawater (saltwater): Seawater is water from the ocean and has a high salinity level.

- It typically contains around 35 ppt salinity, although this can vary depending on factors such as location, depth, and climate.
- Seawater is saline due to the accumulation of dissolved salts, primarily sodium chloride (table salt), along with other minerals and trace elements.

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D. LEVELS OF SALINITY

1. Freshwater culture
2. Mari-culture
3. Brackish water culture





